

Page 4: Project Planning Phase

1. Introduction

The **Project Planning Phase** defines the execution roadmap, resource allocation, task distribution, project timeline, and risk management strategies required to deliver **Smart Budget AI** efficiently and on schedule.

2. Work Breakdown Structure (WBS)

- **1. Initiative Setup & Requirements**
 - 1.1 Project Kickoff & Stakeholder Alignment
 - 1.2 Finalizing SRS (Software Requirements Specification)
- **2. Design & Architecture**
 - 2.1 UI/UX Wireframing & Interactive Prototyping
 - 2.2 System & Database Architecture Setup
- **3. Core Development**
 - 3.1 Backend API & Database Implementation
 - 3.2 Frontend Mobile / Web Application Interface
 - 3.3 AI / ML Pipeline Development (OCR, NLP Categorization, Time-Series Models)
- **4. Testing & Quality Assurance**
 - 4.1 Unit & Integration Testing
 - 4.2 Security, Data Privacy, and Performance Testing
- **5. Deployment & Release**
 - 5.1 Cloud Infrastructure Provisioning & CI/CD Pipeline
 - 5.2 Final User Acceptance Testing (UAT) & Launch

3. Project Schedule & Milestone Timeline (Gantt Chart Representation)

Phase / Task	Duration	Estimated Timeline	Key Deliverables
Phase 1: Requirements & Design	Weeks 1 – 2	Month 1	Approved Wireframes & Architecture Blueprint
Phase 2: Database & Backend Core	Weeks 3 – 5	Month 1 – 2	Functional APIs, Database Schema, Auth System
Phase 3: AI Engine Integration	Weeks 6 – 8	Month 2	Trained NLP & OCR Pipeline Models
Phase 4: Frontend Development	Weeks 7 – 10	Month 2 – 3	Fully Integrated Mobile / Web UI
Phase 5: System Testing & QA	Weeks 11 – 12	Month 3	Bug-Fix Reports, Test Execution Logs
Phase 6: Deployment	Weeks 13 – 14	Month 4	Live Production

Phase / Task	Duration	Estimated Timeline	Key Deliverables
& Launch			Environment & Documentation

4. Resource & Team Allocation

Role	Key Responsibilities	Primary Tools / Tech
Project Manager	Agile sprint management, timeline tracking, resource coordination	Jira, Trello, Notion
UI/UX Designer	User journey mapping, interactive prototypes, visual assets	Figma, Adobe XD
Frontend Developer	Mobile and web interface building, API integration	React Native, Flutter, React.js
Backend Developer	Server architecture, API Gateway, Database queries	Python (FastAPI/Django), PostgreSQL, Redis
AI / ML Engineer	Model training, OCR engine, NLP categorization, LLM tuning	TensorFlow, PyTorch, OpenAI API, Scikit-learn
QA / Testing Specialist	Automated test scripts, manual validation, vulnerability testing	Postman, Selenium, PyTest

5. Project Risk Management Plan

- **Schedule Slippage:** Managed through 2-week Agile sprint cycles and weekly progress standups.
- **Scope Creep:** Managed by maintaining a strict backlog for post-launch releases (V2 features).
- **Resource Bottlenecks:** Mitigated through cross-functional team pairing and cloud sandbox development environments.